

Report

Of

Paytm Clone Application

(Full-Stack Digital Wallet and Payment Application)

Submitted

To

School of Computer Science and Engineering IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

Roll Number of Student: CS - 2341044

Name of Student: AKSHAT VERMA

Batch: 2026-2027

Candidate's Declaration

I, Akshat Verma, hereby declare that the internship report titled *"Paytm Clone Application"* has been prepared by me as part of the requirements for the Bachelor of Technology degree in Computer Science and Engineering.

This report describes the work undertaken during my internship at Heloix Startup Minds Private Limited, including the development and testing of the Paytm Clone Application. The information presented reflects my work and learning during the internship. All external sources used in preparing this report have been appropriately acknowledged.

I further declare that this report has not been submitted to any other institution for the award of a degree or diploma.

Signature

Student Name – AKSHAT VERMA

Roll No. – CS-2341044

Acknowledgement

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism, and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I am thankful to everyone who guided and supported me throughout the internship. I would also like to thank my university faculty for their academic guidance and my family for their continued encouragement.

Signature

Student Name – AKSHAT VERMA

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HELOIX

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INTERNSHIP COMPLETION CERTIFICATE

Date:

TO WHOMSOEVER IT MAY CONCERN

Certificate No.: HELOIX/2026/6044

This is to certify that **Akshat Verma** has successfully completed the internship as a AI Prompt Engineer & Tester with Heloix Startup Minds Private Limited in Work From Home mode from 13/07/2026 to 11/09/2026.

During the internship, Akshat Verma carried out the assigned responsibilities with sincerity, professionalism, and a positive attitude. We appreciate the efforts and contribution made to the organization during this period.

We wish Akshat Verma every success in future academic and professional endeavors.



Authorized Signatory

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4. Project Description

4.1 Introduction

The **Paytm Clone Customer Application** is a full-stack, responsive financial technology (Fintech) web application developed to emulate the end-to-end capabilities of an enterprise-grade digital payment and customer wallet ecosystem. Built to align with the core functional specifications of India's Unified Payments Interface (UPI) and the Bharat Bill Payment System (BBPS), the application empowers users to seamlessly manage digital money, execute peer-to-peer (P2P) and peer-to-merchant (P2M) transactions, link multiple bank accounts, settle utility bills, verify government-issued identity credentials (Aadhaar & PAN), and explore wealth investments and travel services.

The system is developed using modern web engineering principles, pairing a **React 18 Single-Page Application (Vite)** with a modular **Node.js** backend. To guarantee resilience and seamless portability, the database architecture supports a dual-mode pattern: communicating natively with a **PostgreSQL relational database** while providing a deterministic in-memory mock engine fallback for zero-dependency demonstration and offline development. The project showcases industry best practices in client-side state management, double-entry transactional accounting, regulatory compliance handling, and RESTful service design.

4.2 Organisation Profile

[Provide a brief profile of the organisation where the internship was undertaken – name, location, domain of business, year of establishment, size of the organisation, and the department in which the internship was carried out. This section is organisation-specific and must be filled in by the student.]

- Organisation Name: Heloix Startup Minds Private Limited
- Location: Work From Home
- Domain: AI Prompt Engineer & Tester
- Duration of Internship: 2-MONTH

4.3 Problem Statement

Modern digital banking and mobile payment applications require high availability, strict transaction atomicity, low latency, and adherence to regulatory constraints (such as Reserve Bank of India / NPCI mandates on Know Your Customer [KYC] and Merchant classification). Building a comprehensive digital wallet requires solving several engineering challenges

4.4 Project Objectives

The primary objectives executed during the internship include:

- Stateless Multi-Factor Authentication & MPIN Engine: Implement mobile OTP onboarding, encrypted 4-digit MPIN validation, and device fingerprint simulation with JWT session tokens.
- Double-Entry Multi-Bucket Wallet Architecture: Develop an atomic balance management system segregating main wallet balance, cashback balance, and refund credits with an immutable transaction ledger.
- UPI & QR Code Ecosystem: Implement dynamic QR code generation, HTML5 camera-based QR scanning, contact-based UPI transfers, and payment source selection (Wallet vs. Linked Banks).
- Integrated BBPS Utility Payment Gateway: Build a biller discovery and settlement engine supporting telecom recharges, electricity boards, FASTag, DTH, and broadband providers.
- Regulatory eKYC & Role Switching: Establish Aadhaar/PAN verification workflows with role-based permission locks enforcing RBI guidelines for Merchant upgrades.
- Wealth & Travel Micro-Services: Deliver investment simulations (mutual funds, gold, equity) and flight search/booking with automated PNR generation and PDF-style transaction invoices.

4.5 Scope of the Project

The project scope is focused on the Customer Application for digital payments and financial management in the Indian financial context (INR, ₹). It encompasses:

- User registration, OTP login, MPIN verification, and session persistence.
- In-app KYC status tracking (Pending \ Min KYC \ Full KYC).
- Wallet management (Add money via Netbanking/Card/UPI, withdraw to bank, transaction filters).
- UPI payments via VPA ID (user@paytm) and live camera QR code scanner.
- BBPS bill payments with dynamic receipt and transaction record generation.
- Rewards & gamification engine with interactive scratch cards and instant cashback crediting.
- Support ticketing mechanism with automatic ticket identifier generation.

Out of Scope for this phase: Live banking core integrations (NPCI switch / production Bank TSPs), SMS gateway hardware, and payment gateway hardware soundboxes.

4.6 Technologies and Tools Used

The project follows a modern full-stack JavaScript architecture, with the technology choices summarised in Table 4.1.

4.7 System Architecture

4.7.1 High-Level Architecture

The system adopts a classic **Three-Tier Client-Server Architecture** ensuring clear separation of presentation, business logic, and persistent storage: Table 4.1: Technology Stack used in the Project

4.7.3 REST API Documentation

The backend exposes the RESTful endpoints

4.8 Methodology

4.8.1 Development Approach

An **Agile Iterative Lifecycle** was used throughout development:

1. **Sprint 1 (Architecture & Database):** Schema design (schema.sql), seed fixtures (seed.sql), and dual-mode data connector (db.js).
2. **Sprint 2 (Authentication & Core Ledger):** OTP login flows, MPIN validation modal, and multi-wallet balance ledger.
3. **Sprint 3 (Payment Channels & BBPS):** UPI direct transfers, QR scanner modal (html5-qrcode), and bill payment categories.
4. **Sprint 4 (Value-Added Services & Compliance):** Aadhaar/PAN KYC integration, RBI role validator, flight booking, wealth management, and digital invoice generation.
5. **Sprint 5 (Hardening & UI Refinement):** Responsiveness, error boundaries, and integration testing

4.8.2 Authentication, Security & RBI Role Validation Flow

Security and regulatory compliance are enforced at multiple levels:

1. **Multi-Factor Onboarding:** Phone number discovery followed by an OTP handshake and MPIN creation.
2. **Double Authorization Barrier:** High-risk actions (withdrawing money, executing UPI payments, paying bills, and buying investments) enforce interactive **4-digit MPIN validation** via a dedicated modal component before sending the request.

3. **RBI KYC Upgrade Enforcement:** When a user requests an upgrade from a CONSUMER account to a MERCHANT account:
    - The server evaluates mockDb.user.kyc_status === 'FULL_KYC'.
    - If the user only holds MIN_KYC or PENDING, the server blocks the request with an **HTTP 403 Forbidden** status:
      "⚠ RBI Norms: Complete Full Aadhaar & PAN eKYC before upgrading to Merchant Account."
    - Only after completing biometric/OTP Aadhaar and PAN verification is the merchant feature set unlocked.

4.8.3 Smart Financial Insights Engine

To prevent phantom balances and ensure auditability, transactions follow strict ledger rules:

Table 4.3: Financial Transaction Rules & Validation Checks

4.8.4 Testing Strategy

- **Unit Testing:** Verified core mathematical functions including ledger balance calculations, discount computations, and input masking (Aadhaar XXXX-XXXX-4920).
- **API Integration Testing:** Validated all endpoints across positive and negative paths using REST payloads (e.g., validating invalid MPIN responses, testing 403 role lockouts, and verifying wallet balance deductions).
- **Hardware & Camera Testing:** Tested the html5-qrcode library across various lighting conditions, ensuring camera access handshakes and QR string parsing work reliably.
- **Responsive UI Verification:** Verified layout compatibility across mobile viewports (360px–420px), tablets (768px), and desktop displays (1080p).

4.9 Expected Outcomes

The completed Paytm Clone application achieved all planned milestones:

- **Fully Functional Digital Wallet:** Users can add money, withdraw to linked accounts, and track balances across three distinct wallet accounts.
- **Fast UPI & QR Experience:** Real-time QR code generation and camera scanning allow users to complete payments in seconds.
- **Regulatory Rigor:** Demonstrated strict implementation of RBI eKYC compliance checks before granting merchant privileges.

- **Rich Utility Suite:** Unified mobile recharges, electricity payments, flight bookings, and investment purchases into a single cohesive interface.

- **Zero-Setup Demonstration:** The dual-mode database pattern allows the entire system to run out-of-the-box using the built-in mock engine or scale to a live PostgreSQL database.

4.10 Certificates of Completion and Communication Proof

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